

CHANGELOG

Changelog

All notable changes to this project will be documented in this file.

The format is based on [Keep a Changelog](#), and this project adheres to [Semantic Versioning](#).

[Unreleased]

Added

- **Foundation distributed-systems library (pkg/)** — a large, growing pure-Go, deterministic, mutation-tested package set. Recent additions include:
 - Consensus/coordination: voting, failconfirm, heartbeatcoalescer, splitbrainalert.
 - Replication/state: mvcc (B-tree time-travel store), antientropy (hinted-handoff + read-repair + Merkle diff), watchmanager (syncd/unsyncd/victim watch groups), crdt/merkle.
 - Scheduling/placement: constraints (Pacemaker 4-type), preempt (value-multiplier), priorityqueue (multifactor aging), workclaim (SKIP-LOCKED), admissioncontrol (N+K reserve), budgetcap, qos, suitability, ewmarank, fallbackchain.
 - GPU/resource & cost: costsched, latencysched, healthmonitor, gpuattest (attestation crypto: challenge/response, proof-of-GPU-work, O(1) spot-check, device-sealing), attestadmit, quantization, local (TCO), pool, burst.
 - Messaging/flow & edge: flowcontrol (K8s APF), idempotent (exactly-once), rebalance (cooperative-sticky), informer (list-watch cache), redundantexec (BOINC trust), edgeregistry, edgeverify, slotmigration, hashslot, healthprobe.
 - Testing/simulation: timefault (clock skew/freeze/monotonic injectors), BUGGIFY in testing/dst, phase7matrix (gap-matrix verifier).
 - This session (34 pure-Go units):
 - bursthysteresis — BurstController with a MONITOR→SPILL→RECOVER two-threshold hysteresis dead-band (HXC-1504).
 - providerchain — ordered multi-tier provider fallback cascade with retrievable/terminal error classification and injected-clock failover SLA (HXC-1508).

- `gpucatalog` — `SUPPORTED_GPUS` compute-multiplier catalog, `LookupMultiplier`, and attestation-aware Score (attested above un-attested) (HXC-1592).
- `gravaladmit` — HMAC-SHA256 GraVal challenge-response provider admission gate with single-use nonce and `graval.verified` label (HXC-1523).
- `modelrouter` — strategy-to-default-model resolution (latency/throughput/quality/cost) with a typed unknown-strategy error (HXC-1430).
- `gravalverify` — GraVal GPU attestation with a VRAM-ratio ≥ 0.95 gate and a concurrent, race-clean `BatchVerify` pass-rate KPI (HXC-1435, HXC-1436).
- `gepetto` — HelixGepetto dual-resource local-vs-Chutes arbitration with a monotonic reserve schedule and a strict >0.80 high-water anti-starvation zero-capacity cut-off (HXC-1446).
- `chutesaccount` — Chutes API model-list (TEE/price fields) and /users/me account-balance client over HTTP with `Authorization: Bearer` (HXC-1429).
- `cmd/burst-controller` — utilization-driven binary driving `bursthysteresis`; emits a “burst allocation request” on `SPILL` and a `RECOVER` line, `runID`-stamped (HXC-1531).
- `tieredcache` — hot(memory)/warm(NVMe)/cold(SSD) tiered cache with injected backends + injected clock, idle-demotion + promotion, and per-tier hit stats (HXC-1418).
- `smartrouter` — intelligent `ModelRouter` (latency/throughput/cost/TEE/balanced) that always excludes unhealthy models, with a typed no-eligible-model error (HXC-1539).
- `epochresolve` — config-epoch failover slot-ownership conflict resolution (highest-epoch wins, deterministic lexicographic equal-epoch tiebreak, order-independent convergence) (HXC-1397).
- `balancemonitor` — USD account-balance monitor with a strict low-balance floor warning, over HTTP with `Authorization: Bearer` (HXC-1512).
- `scan` — Oracle-SCAN-style stable virtual client endpoint: least-loaded healthy-backend routing with a membership-stable name and injected clock (HXC-1403).
- `llmfailover` — error-classified LLM failover taxonomy (deterministic per-class fallback paths) plus a sandbox scheduling-hint contract (HXC-1600).
- `llmadapter` — Claude Messages / OpenAI Responses-API request and response shape adapters to OpenAI chat-completions, preserving tool-call order (HXC-1599).
- `deltacrdt` — standalone delta-state CRDTs (G-counter / PN-counter / observed-remove OR-set / LWW-map) with delta-mutators and a convergent (commutative/associative/idempotent) merge (HXC-1409).
- `gputopo` — topology-aware NUMA/NVLink GPU placement scoring that prefers NVLink-connected GPU pairs and falls back to PCIe (HXC-1406).
- `fsresidency` — filesystem residency challenge: per-byte-range `ReadAt` + SHA-256 verification proving a model file is really present on disk; truncated/missing/mismatch all fail (HXC-1572).
- `economics` — `RewardDistributor` multi-token distribution with treasury/reinvest splits and a conservation invariant (`sum(ledger) + treasury + reinvest == total`), plus `GetParticipantROI` (ROI%

+ break-even days, typed unknown-participant error) (HXC-1460, HXC-1461).

- `revenueopt` — `RevenueOptimizer.OptimizeAllocation` greedy GPU→marketplace assignment maximising expected revenue, with a TEE→Chutes revenue bias; self-contained types (HXC-1456).
- `exportcontrol` — export-control / country-code KYC gate: a controlled GPU (H100/A100) from a Tier-3 (embargoed) or unknown jurisdiction is rejected with `ErrExportDenied`; Tier-1 is allowed (HXC-1482).
- `devicecatalog` — machine-readable 64-device taxonomy with case-insensitive substring Lookup resolving a discovered CPU model (e.g. Rockchip RK3588) to its tier / trust-level / compute-class (HXC-1331).
- `compliancecdoc` — EU AI Act compliance-documentation pipeline: generates a model card + technical-documentation artifact from attestation-log records, with a SHA-256 provenance hash bound to the actual record content (HXC-1481).
- `hybridkex` — hybrid post-quantum key exchange combining X25519 (crypto/ecdh) and ML-KEM-768 (crypto/mlkem) shared secrets via HKDF-SHA256 (group X25519MLKEM768); a tampered ML-KEM ciphertext yields a non-matching key (HXC-1476).
- `carbonsched` — carbon-aware job placement: chooses the lowest-carbon-intensity latency-eligible region (deterministic name tiebreak) with per-job kWh / gCO2 metering (HXC-1480).
- `gpuattest` (additive `multigpu.go`) — multi-GPU node enumeration: an N-GPU node descriptor yields N distinct, independently-verifiable per-device attestation proofs keyed by GPU UUID; rejects duplicate/empty descriptors (HXC-1573).
- `workloadrouter` — `UnifiedManager.RouteWorkload`: concurrent per-marketplace price probes + a weighted composite score (inverse price/latency, health) with a TEE multiplier that re-ranks confidential offers when the workload requires a TEE (HXC-1455).
- `metrics` (additive `tiermetrics.go`) — `TierMetrics` exposes `gpu_tier_utilization`, `gpu_cost_per_hour`, and `provider_health` Prometheus-text series with deterministic ordering and concurrent-safe setters (HXC-1535).
- `cmd/gpu-pool-manager` — GPU pool manager HTTP front-end: `POST /allocate` returns the tier-correct (lowest-tier healthy, name tiebreak) provider as JSON (503 when none healthy) and `GET /providers` lists providers with health; `httptest-gated`, self-contained (HXC-1530).
- `marketplaceadapter` — `MarketplaceAdapter` interface (`Name/GetCurrentPricing/SubmitWork`) with a `Name()`-keyed dispatch Registry and a `ChutesAdapter` over HTTP (`httptest` fixture, non-2xx rejected) (HXC-1454).
- `cmd/e2ee-proxy` — transparent ML-KEM-768 (crypto/mlkem) + AES-256-GCM encrypting proxy: `EncryptingTransport` seals request bodies and `DecryptingHandler` opens them, so the on-wire payload between proxy and upstream is ciphertext-only while the client receives the correct plaintext; tampered ciphertext is rejected (HXC-1532).
- `e2eebench` — `MeasureHandshakeLatency` benchmarks the `hybridkex` full handshake (median + P95), proving median < 1ms on

host and enforcing per-iteration key agreement (ErrKeyDisagreement) (HXC-1556).

- `archlint + docs/ARCHITECTURE.md` — the hardened L0–L7 architecture diagram and component map, mechanically enforced: `archlint` parses the component-map table and fails the build if any documented component maps to a package path that does not exist on disk (HXC-1424).
- Wave 67 (5 units, subagent-driven, 5 parallel streams in disjoint packages):
 - `multiraft` — `MultiRaftManager` partitioning cluster state into independent per-shard `go.etcd.io/raft/v3` groups, each electing its own leader and committing independently via per-shard `Propose` routing over an in-process `RaftTransport`; write throughput scales with shard count and a shard re-elects on leader loss while others keep committing (HXC-1387).
 - `stonith` — STONITH fencing: a `FencingAgent` interface with IPMI / AWS-EC2 / Azure-ARM / SBD-shared-disk / NoOp drivers and a `MultiLevelFencer` multi-level fallback, with sink-side fence confirmation; the IPMI real-binary path is an honest SKIP-with-reason when `ipmitool` is absent (HXC-1401).
 - `deviceplugin` — Nomad/K8s-style gRPC device-plugin / GRES fingerprinting framework over real gRPC: plugins register full device fingerprints, the registry parses GRES descriptors, atomically applies fingerprints, allocates by request, and rejects oversubscription (HXC-1407).
 - `internal/gpu` (additive `manager_helixpow.go`) — dual-workload capacity reservation: `ReserveForHelixPoW(fraction) + ChutesCapacity()` reporting only the non-reserved remainder, with a strict >0.80 Helix-load Gepetto starvation guard that zeroes Chutes capacity (HXC-1447).
 - `chutes` (additive `config.go`) — `ChutesMinerConfig` and `ValidatorConfig` deployment descriptors with `Validate()` invariants (non-empty IDs/hotkeys, `GPUCount > 0`, non-negative cost, positive cache size, TEE consistency) (HXC-1439).
- Wave 68 (5 units, subagent-driven, 5 parallel streams in disjoint packages, run concurrently with wave 69):
 - `dst` — standalone FoundationDB-style deterministic simulation harness: single-threaded seeded-RNG event loop with injectable network/disk/logical-clock; same seed $\Rightarrow$ byte-for-byte identical trace + exact failure replay; 1000+ seeded sims (HXC-1419).
 - `porcupine` — self-contained WGL linearizability checker + concurrent history recorder: PASS on linearizable, FAIL (with violating op) on a seeded non-linearizable history (HXC-1421).
 - `kraft` — KRaft-style self-managed Raft metadata quorum (`go.etcd.io/raft/v3`, no ZooKeeper): replicated `CreateTopic/ListTopics` + partition-leader assignment, controller election + controller-loss failover (HXC-1417).
 - `internal/chaos` — fault-injection library (`PodKill/NetworkPartition/DiskStall/ClockSkew`) + canary-guarded runner with automatic rollback on `health < SLO`; deterministic via injected clock + seeded selection (HXC-1422).
 - `multiraft` (additive `lease.go`) — `LeaseTracker` CockroachDB-style leaseholder local reads (no Raft round-trip), follower routing via `RaftTransport.SendRead`, lease-expiry re-route against an injected clock (HXC-1389).

- Wave 69 (5 units, subagent-driven, 5 parallel streams in disjoint packages, run concurrently with wave 68):
 - stonith — IPMI credential hardening: BMC password delivered via the IPMI_PASSWORD env + -E (never on the argv / process table) with redacted argv in error strings; tests assert the password appears in neither argv nor errors (HXC-908, closing a wave-67 follow-up).
 - deviceplugin — concurrency test driving ApplyFingerprint/Allocate/Release from 50+ goroutines under -race, proving the Registry never oversubscribes (mutex removal trips the race detector) (HXC-910, closing a wave-67 follow-up).
 - internal/llm — replaced the stub Inference with a real strategy-based router (latency/throughput/quality/cost) over the live model registry, excluding unhealthy/unloaded models with a typed no-eligible-model error and a real injectable Backend seam (HXC-1470).
 - metrics (additive earningsmetrics.go) — EarningsMetrics exposing tao_earnings_total / graval_attestation_status / token_throughput / gpu_utilization Prometheus-text series with deterministic (sorted) ordering + concurrent-safe setters (HXC-1483).
 - internal/gpu (additive attesthook.go) — GraVal attestation hook: AllocateForChutes refuses a GPU that has not passed attestation (ErrAttestationRequired), even on the idempotent re-allocation path; injectable ChutesAttestor seam (HXC-1448).
- Wave 70 (5 units, subagent-driven, 5 parallel streams in disjoint packages, run concurrently with wave 71):
 - provider/chutes — OpenAI-compatible /v1/chat/completions provider (cpk_Bearer) with HTTP-429 retry+backoff; httpstest proves retry-then-success, Bearer header, and typed errors (HXC-1510).
 - pool (additive model.go) — GPUDevice/WorkloadSpec/GPUAllocation/PoolStatus data model with snake_case JSON tags + round-trip/concurrency tests (HXC-1495).
 - quantization (additive awq.go) — AWQ 4-bit default serving format + EstimateVRAM (~25% of fp16) + budget-aware SelectFormat preferring AWQ when fp16 won't fit (HXC-1473).
 - internal/gateway (additive inference.go) — inference route forwarding to an injectable Chutes client; TEERequired => X-E2EE-Enabled routing marker; client failure => 502 (HXC-1469).
 - internal/gpu (additive mig.go) — MIG profile management: standardized vGPU tiers (1g.10gb/2g.20gb/3g.40gb/7g.80gb), in-memory partitioning with oversubscription rejection + fixed-at-creation immutability (HXC-1449).
- Wave 71 (5 units, subagent-driven, 5 parallel streams in disjoint packages, run concurrently with wave 70; closes 4 prior follow-ups):
 - multiraft — RaftTransport async-delivery safety: inbox handler Steps under the shard mutex, outbound messages deferred + flushed after unlock; a new async-transport -race test proves a goroutine-delivering transport is race-free and still commits (HXC-909).
 - kraft — CreateTopic returns ErrTopicExists on a conflicting re-create (different partition count); identical re-create stays idempotent (HXC-912).
 - porcupine — WGL state-dedup memoization via Model.Equal: verdict-preserving pruning (memo fires on backtracking histories; identical PASS/FAIL + witness vs Equal==nil) (HXC-913).

- chutes — StreamChannel honors ctx cancellation on a blocked read (prompt channel close, no goroutine leak) and closes the owned reader (HXC-911).
- marketplaceadapter (additive akash.go) — Akash (Cosmos/AKT) adapter over an injectable client: reverse-auction pricing, SDL submission, provider-reputation gate, registered in the Name()-dispatch Registry (HXC-1458).
- Wave 72 (5 units, subagent-driven, 5 parallel streams in disjoint packages):
 - provider/runpod — RunPodProvider (implements pool.GPUProvider) with a warm-pool fast path over an injectable client (HXC-1515).
 - provider/aws — AWSProvider EC2 Spot adapter over an injectable EC2-client interface (no aws-sdk dep): GPU model→instance-type selection (p5/p4de/g5), Spot+tags, TOCTOU-safe capacity gate (HXC-1516).
 - scheduler (additive tier_filter.go) — GPURTier-aware filter predicate: hard-exclude disallowed tiers + preference scoring, composes with the existing count filter, fails closed on a nil predicate (HXC-1534).
 - provider/chutes — ctx-interruptible 429 backoff (select vs ctx.Done()) + honors Retry-After (delta-seconds/HTTP-date, max(hint,backoff)) (HXC-916, closing a wave-70 follow-up).
 - grafanadash (additive tiercost_dash.go) — Phase-8B tier-utilization + cost dashboard generator (panels target gpu_tier_utilization/gpu_cost_per_hour), valid + deterministic (HXC-1551).
- **Security fix (HIGH):** closed a TOCTOU replay-bypass in pkg/gpuattest.Verify — the nonce check and consume are now atomic, with a concurrent-replay -race regression test.
- Every foundation package ships with tests that fail under mutation of the logic they cover (CLAUDE-1 enforcement) and are gated by whole-tree build/vet + -race.
- Initial project scaffold with Go workspace.
- 29 submodules integrated at project root.
- Directory structure: cmd/, pkg/, internal/, api/, web/, scripts/, deploy/, test/.

Changed

- **Governance:** CLAUDE-3 / AGENT-2 / QWEN-2 documentation continuous-sync guarantee added (restates and cites Constitution §11.4.106 docs-chain); .docs_chain/contexts/tracked_docs.yaml extended to enforce README, CLAUDE/AGENTS/QWEN, CHANGELOG, FOUNDATION_PACKAGES, and CODEGRAPH export-sync.

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Added

- Phase 0 bootstrap: submodule cloning and workspace initialization.